Self Learning – Programs based on dictionary

1. Student Marks Analysis

Sample Data

```
marks = {
  "Aarav": 78,
  "Diya": 92,
  "Rohan": 45,
  "Ishita": 88,
  "Kabir": 56,
  "Meera": 39,
  "Arjun": 95,
  "Saanvi": 67,
  "Vivaan": 82,
  "Anaya": 51
}
```

Tasks

- Display students scoring **80 or above**.
- Count the number of students who **failed** (marks < 40).
- Find the **highest scorer**.
- Create a list of students scoring between **60 and 75**.
- Assign grades:
  - A: ≥ 90
  - B: 75–89
  - C: 50–74
  - F: < 50

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2. Inventory Management System

Sample Data

```
inventory = {
  "Notebook": 45,
  "Pen": 120,
  "Pencil": 80,
  "Eraser": 25,
  "Marker": 15,
  "Stapler": 8,
  "Glue": 12,
  "Scale": 30,
  "Folder": 5,
  "Calculator": 3
}
```

Tasks

- Display products with stock **less than 10**.

- Count products having stock **more than 50**.
- Find the product with the **minimum stock**.
- Create a list of products that require **restocking** (stock < 20).
- Calculate the **total inventory count**.

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3. Employee Salary Processing

Sample Data

```
salary = {
  "EMP101": 45000,
  "EMP102": 62000,
  "EMP103": 38000,
  "EMP104": 75000,
  "EMP105": 54000,
  "EMP106": 29000,
  "EMP107": 82000,
  "EMP108": 48000,
  "EMP109": 36000,
  "EMP110": 68000
}
```

Tasks

- Display employees earning above **₹60,000**.
- Count employees earning below **₹40,000**.
- Find the **highest-paid employee**.
- Create a list of employees eligible for a **bonus** (salary > ₹50,000).
- Calculate the **average salary**.

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4. Cricket Scoreboard Analysis

Sample Data

```
scores = {
  "Virat": 78,
  "Rohit": 112,
  "Gill": 45,
  "Rahul": 89,
  "Hardik": 32,
  "Jadeja": 61,
  "Surya": 105,
  "Pant": 95,
  "Bumrah": 18,
  "Shami": 25
}
```

Tasks

- Display players who scored **50 or more runs**.

- Count the number of **centuries**.
- Find the player with the **highest score**.
- Create a list of players scoring below **30 runs**.
- Determine how many players scored between **50 and 99**.

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5. Library Book Availability

Sample Data

```
books = {
  "Python Basics": 5,
  "Data Structures": 0,
  "Machine Learning": 3,
  "Java Programming": 2,
  "DBMS": 0,
  "Operating Systems": 6,
  "Networking": 4,
  "Cloud Computing": 1,
  "Cyber Security": 0,
  "Web Development": 7
}
```

Tasks

- Display books that are **currently unavailable**.
- Count the number of available books.
- Find the book with the **maximum copies**.
- Create a list of books having **less than 3 copies**.
- Calculate the total number of books available.

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6. Product Price Analysis

Sample Data

```
prices = {
  "Laptop": 55000,
  "Mouse": 800,
  "Keyboard": 1800,
  "Monitor": 12000,
  "Printer": 9000,
  "Tablet": 28000,
  "Speaker": 3500,
  "Webcam": 2500,
  "Headphones": 4200,
  "Router": 3200
}
```

Tasks

- Display products costing more than **₹5000**.

- Count products costing less than **₹3000**.
- Find the **most expensive product**.
- Create a list of products priced between **₹2000 and ₹10000**.
- Calculate the total value of all products.

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7. Bus Route Passenger Analysis

Sample Data

```
passengers = {
  "Stop1": 12,
  "Stop2": 25,
  "Stop3": 18,
  "Stop4": 32,
  "Stop5": 9,
  "Stop6": 28,
  "Stop7": 14,
  "Stop8": 7,
  "Stop9": 21,
  "Stop10": 16
}
```

Tasks

- Display stops having more than **20 passengers**.
- Count stops with fewer than **10 passengers**.
- Find the **busiest stop**.
- Create a list of stops requiring an **extra bus** (passengers > 25).
- Calculate the average number of passengers.

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8. Online Quiz Performance

Sample Data

```
quiz_scores = {
  "S001": 18,
  "S002": 12,
  "S003": 9,
  "S004": 20,
  "S005": 14,
  "S006": 7,
  "S007": 16,
  "S008": 10,
  "S009": 19,
  "S010": 13
}
```

(Quiz is out of 20 marks.)

Tasks

- Display students scoring **15 or above**.
- Count students scoring below **10**.
- Find the **top performer**.
- Create a list of students who passed (**≥ 10 marks**).
- Calculate the class average.

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9. Mobile Contact Directory

Sample Data

```
contacts = {
  "Amit": "9876543210",
  "Priya": "9876543211",
  "Rohan": "9876543212",
  "Neha": "9876543213",
  "Anjali": "9876543214",
  "Karan": "9876543215",
  "Pooja": "9876543216",
  "Arjun": "9876543217",
  "Sneha": "9876543218",
  "Rahul": "9876543219"
}
```

Tasks

- Display all contact names in **alphabetical order**.
- Count the total number of contacts.
- Search for a given contact name.
- Create a list of contacts whose names start with a vowel.
- Stop the search using break once the required contact is found.

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10. Electricity Consumption Report

Sample Data

```
units = {
  "House101": 320,
  "House102": 180,
  "House103": 450,
  "House104": 290,
  "House105": 150,
  "House106": 510,
  "House107": 220,
  "House108": 390,
  "House109": 170,
  "House110": 260
}
```

Tasks

- Display houses consuming more than **300 units**.
- Count houses consuming less than **200 units**.
- Find the house with the **highest consumption**.
- Create a list of houses eligible for an **energy-saving awareness campaign** (consumption > 400 units).
- Categorize houses as:
    - Low: < 200 units
    - Medium: 200–350 units
    - High: > 350 units